Prodigy Anesthesia Quick Review

Femoral Block

Where is the femoral nerve positioned in the femoral canal?	The femoral nerve maintains a lateral position in the femoral canal. In order from lateral to medial the structures are remembered by the acronym NAVEL (nerve, artery, vein, empty space, and lymphatics). *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1003.*
Where is the femoral nerve positioned at the level of the inguinal crease?	At the inguinal crease (perhaps the easiest place to locate it), it is found avout 0.5 cm lateral to the femoral artery. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1634.*
What muscles receive innervation from the femoral nerve?	The posterior branches of the femoral nerve provide motor input to the quadriceps. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 840.*
What area receives sensory innervation from the femoral nerve?	The skin of the anterior thigh from the inguinal ligament to the knee. The terminal branch of the femoral nerve (the saphenous nerve) supplies the medial side of the leg from the knee to the great toe. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1002.*
What nerve roots contribute to the femoral nerve?	L2, L3, L4 *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 840.*
Is a femoral nerve block adequate as the sole anesthetic for an anterior cruciate ligament repair with a hamstring autograft? Why or why not?	No, because the area incised for the hamstring autograft is innervated by the sciatic nerve. A combination of femoral nerve block and sciatic block will be required for this procedure. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1634.*
How is correct needle placement identified for a femoral nerve block when using a nerve stimulator?	Correct needle placement is confirmed when stimulation of the quadriceps and sartorius muscles is seen. (This often causes a 'patellar snap' or upward motion of the patella when the nerve stimulator is set at 0.3-0.5 mA.) *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1634.*

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Femoral Block

How would you block the lateral femoral cutaneous nerve?

The lateral femoral cutaneous nerve emerges underneath the inguinal ligament and becomes superficial inside the fascia lata near the anterior superior iliac spine. It can be anesthetized by inserting a needle 2 cm medial and caudal to the anterior superior iliac spine and injecting 10-15 mL of local anesthetic after the needle penetrates the fascia lata.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1004-1006.

You have performed a three-in-one block on a patient for postoperative pain control. The patient is concerned because he is unable to adduct the affected leg after the performing the block. What is the cause of his inability to adduct the leg?

The obturator nerve provides sensation to the medial aspect of the thigh and motor innervation to the adductor muscles located in the medial thigh. This nerve is anesthetized during a three-in-one block.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1001.

What is a 'three-in-one" block?

A "three-in-one" block anesthetizes the lateral femoral cutaneous, obturator, and femoral nerves and is performed by injecting 20-40 mL of local anesthetic into the neurovascular sheath just lateral to the femoral pulse. The term "three-in-one block" has almost been abandoned in recent years because of the high rate of failure to block all three nerves with a single injection.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1002.

For what procedures is a femoral nerve block useful?

It provides adequate anesthesia for surgeries limited to the anterior thigh, such as muscle biopsies, and has been reported useful for knee arthroscopy and repair of midfemoral fractures.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 840.

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